

Scale invariance between predictions and realizations

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We study the relation between predictions and outcomes. Here prediction is about the occurrence of some event. Predictions are given in terms of probabilities and they are serialized in order of the magnitudes of the probabilities. We study the cumulative functions of the occurrences of the events that predictions foresee. We find scale invariant behavior in a wide variety of events, defaults of companies, entrance examinations, horse races and so on. We point out two mechanisms of the scale invariance between predictions and realizations. In the process, we introduce a simple voting model. The mixing of strong and weak horses is induced by voting and the scale invariant behavior appears automatically. The cumulative function of strong horses x_s and that of the weak horses x_w obeys power law $1 - x_s \sim (1 - x_w)^\alpha$ in the small share of the vote region. In a limit, the relation $1 - x_s = (1 - x_w)^\alpha$ holds exactly over the entire range, which is a remarkable feature of the voting model.

The ubiquity of scale invariant behaviors in the natural world and man-made phenomena is astonishing and even now it is an area that attracts considerable research interest [1]. However, up to now, the measured values which obey power-law distributions is quantitative ones. Here, we discuss the scale invariance which appears in the realizations and non-realizations of predictions. Predictions are serialized in order of the magnitudes of the probabilities and we study the cumulative functions of the occurrence of the events which the predictions foresee. Two plot of the cumulative function is known as Lorenz curve. Figure 1 and Figure 2 plot the Lorenz curves for the default of companies. As the magnitude of the probability whether a company defaults or not, we use the financial scores. The double logarithmic plots of the Lorenz curves become straight which exhibits the scale invariance in the default predictions.

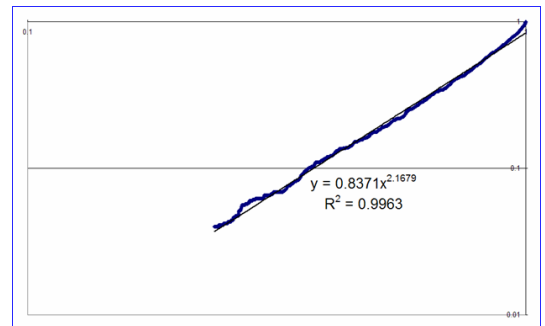
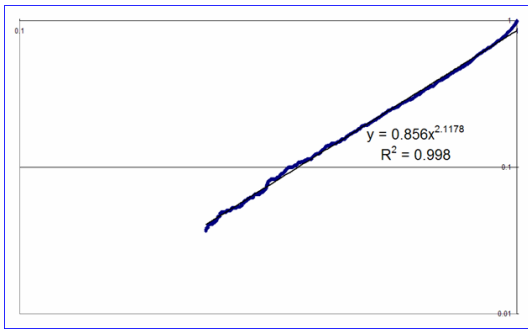


Figure 1: Plot of Lorenz curve for financial scoring model.

Figure 2: Plot of Lorenz curve for EDF model.

Such a scale invariance can be seen in horse races. Here, we consider the share of the votes of

a horse in a race as the probability that the horse wins the race. We denote the cumulative function of the i -th finishing horses as x_i . The two dimensional plot of (x_i, x_j) is known as receiver operating characteristic (ROC) curve. Figure 3 shows the double logarithmic plot of the ROC curves for three combinations of $(i, j) = (1, 2), (1, 11)$ and $(8, 11)$. As in the previous case they become straight, which means that the scale invariant mixing of strong and weak horses is universal.

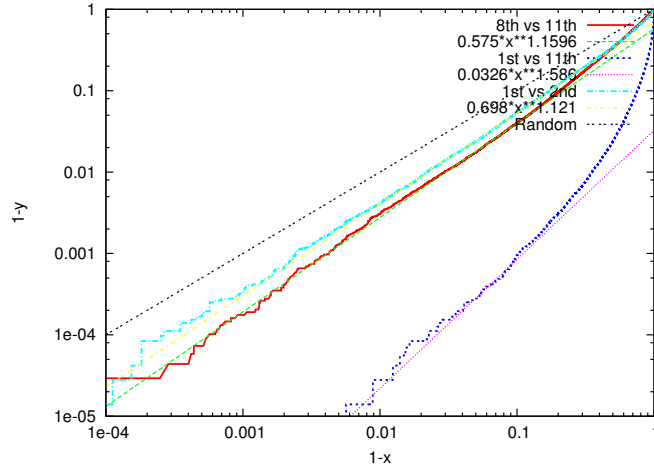


Figure 3: Plot of ROC curve of strong i and weak horses j . We use 71549 horse races of JRA from 1986 to 2006.

We will clarify the mechanism of the scale invariance between predictions and outcomes. In the process, we study a voting model for binary horses which are classified into two classes, strong one and weak one [2]. All betters know the information “partially” and they are encoded in binary scores s and w . Here we use the word “partially”, the reason is that the betters do not have certainty about the information. They decide based on the scores and other information with a probabilistic rule. Even if strong horses have larger score than weak horses, betters do not necessarily vote to good candidates. We find that the mixing of them by voting become scale invariant in the region of the small share of the votes. The cumulative function of strong horse x_s and that of weak horses x_w obey the relation $1 - x_s \propto (1 - x_w)^{s/w}$. In addition, we show the remarkable feature that the scale invariance does hold exactly over the entire range. Taking the zero score limit $s, w \rightarrow 0$ with s/w fixed, the following relation does hold

$$1 - x_s = (1 - x_w)^{s/w}.$$

Keywords

voting, scale invariance, ROC curve, horse race, self-organized

References

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- [2] S. Mori and M. Hisakado, *Exact Scale Invariance in the Mixing of Binary Candidates in a Voting System*, preprint <http://jp.arxiv.org/abs/0806.0185>.